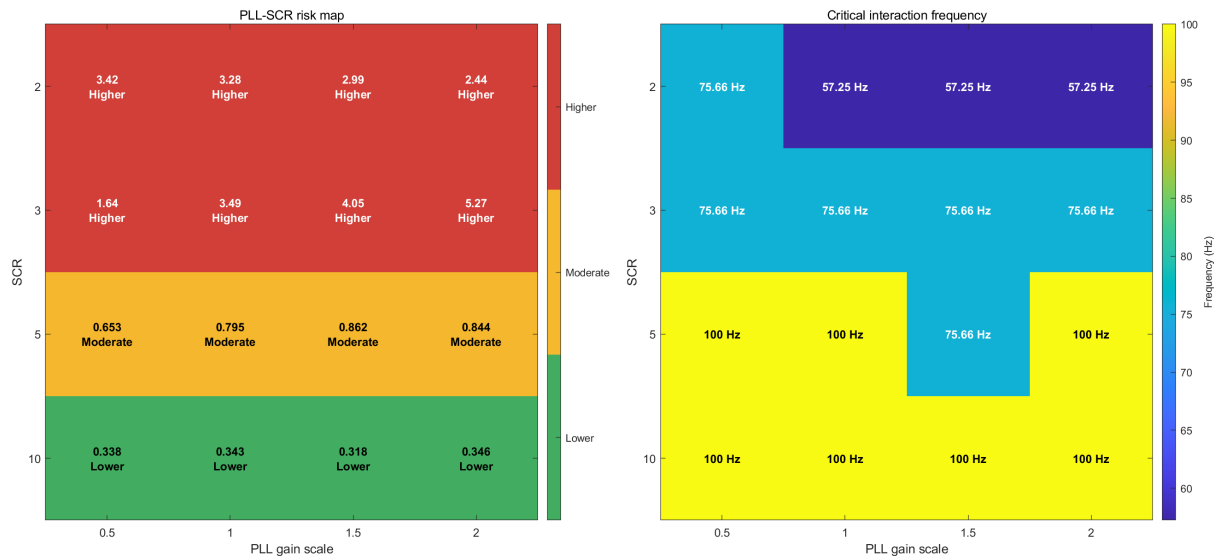


Automated Frequency-Scan and Weak-Grid Stability Screening of a Grid-Following BESS

Black-box dq admittance identification, SCR-PLL sensitivity and time-domain cross-check

Frequency range	Operating matrix	Study volume	Validation
0.5-100 Hz 20 points	4 SCR x 4 PLL 16 cases	640 d/q injection runs	7/7 local checks passed



Executive summary

This project converts repeatable d-axis and q-axis terminal perturbations into a complete black-box dq admittance, combines the result with grid impedance, and screens sixteen SCR-PLL operating cases using one fixed interaction metric. SCR 10 remains Lower, SCR 5 is Moderate, and SCR 3 and 2 are Higher across all PLL scales. Critical frequencies fall between 57.25 and 100 Hz.

The project is a comparative screening tool. It is not a formal stability certificate, grid-code compliance result, or substitute for EMT, HIL, vendor-model or measurement validation.

Henry Tang | Version 1.0 | 2 September 2026

1. Scope and system boundary

The study reuses a validated 11 kV, 50 Hz grid-following BESS containing an LCL filter, average VSC, physical DC link, bidirectional DC/DC stage, RC battery, SoC supervision, PLL, abc/dq conversion, dq current loops, P/Q control, DC-link regulation and protection support. The converter is treated as a black-box terminal device; identification uses PCC voltage and current behavior rather than internal analytical equations.

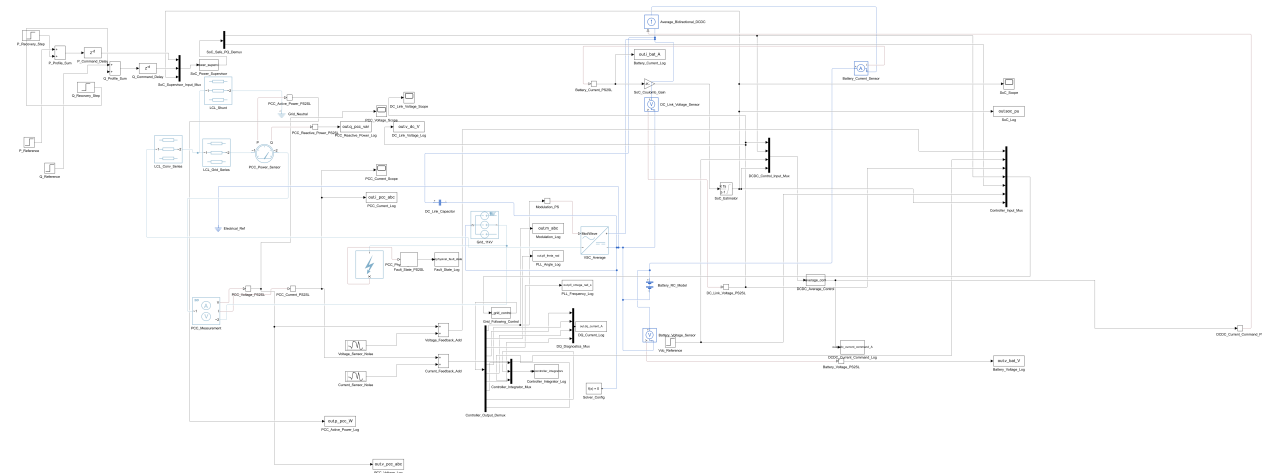


Figure 1. Simulink top-level model used by the frequency-scan study.

Operating point

Quantity	Value	Purpose
Grid	11 kV, 50 Hz	Project baseline
Active power	approximately 0.5 pu	Defined identification point
Reactive reference	0	Neutral baseline
Voltage perturbation	0.3% pu	Measurable local response

When perturbation is disabled, the scan model and source model are run with the same timing. Nine signals match with zero measured numerical error, and the model files are byte-identical at the validated revision.

The perturbation is intentionally small so that the scan does not deliberately activate current limits, SoC boundaries or fault-support logic.

2. Frequency-scan method

Every frequency uses one shared perturbation-disabled baseline, one d-axis injection and one q-axis injection. After the configured initial cycles are discarded, a least-squares sine/cosine fit extracts complex voltage and current tones. The two excitation columns are combined as:

$$\begin{bmatrix} \Delta I_d & \Delta I_q \end{bmatrix}^T = Y_{dq}(j\omega) \begin{bmatrix} \Delta V_d & \Delta V_q \end{bmatrix}^T$$

$$Y_{dq}(j\omega) = I(j\omega) V(j\omega)^{-1}$$

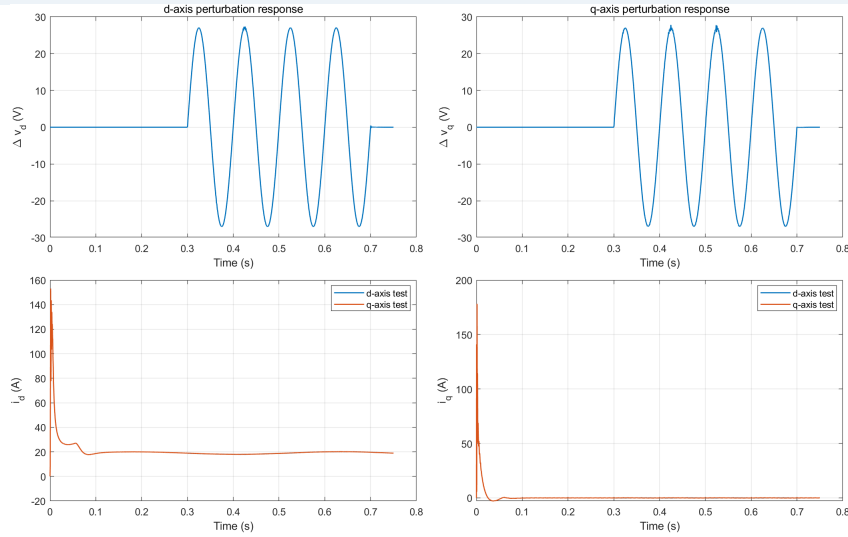


Figure 2. Independent d-axis and q-axis excitation at the 10 Hz reference point.

Why two runs are required

The d-axis run supplies the first response column and the q-axis run supplies the second. Neither is a duplicate. PLL dynamics, rotating-frame terms, plant dynamics and power loops can produce non-zero cross coupling, so Y_{dq} and Y_{qd} are retained.

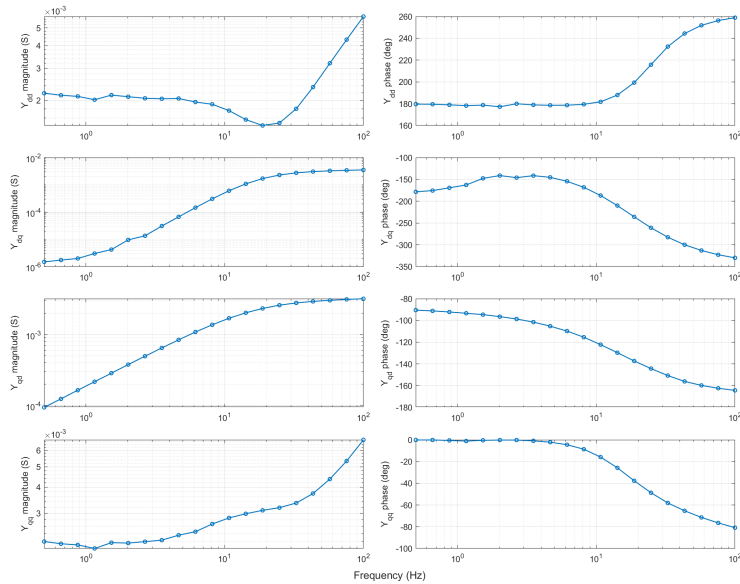


Figure 3. Magnitude and phase of Y_{dd} , Y_{dq} , Y_{qd} and Y_{qq} from 0.5 to 100 Hz.

Completed points are saved incrementally with a configuration signature, allowing interrupted scans to resume without silently reusing data from a different SCR, PLL or quality configuration.

3. Identification quality and repeatability

Formal quality gates replace an informal finite-value check. Near-zero cross-voltage channels are evaluated by leakage relative to the main injected voltage. Current residuals are magnitude-weighted across the full matrix, while the 10 Hz reference point also requires every raw current residual to remain below 35%.

Quantity	Limit	10 Hz	Full band
Main voltage residual	2%	0.58%	0.63%
Weighted current residual	35%	8.84%	14.23%
Raw current residual	35% at 10 Hz	29.66%	99.99% report only
Cross-voltage leakage	5%	0.89%	3.29%
Voltage matrix condition	< 10,000	Pass	Pass

The full-band 99.99% raw current residual occurs in a response phasor close to zero. It is preserved for transparency but does not dominate the acceptance decision. The weighted current metric is the formal full-band gate.

Independent repeatability

The complete 10 Hz d/q identification is run twice in separate simulations and every complex admittance element is compared. Fixed limits are 2% magnitude difference and 2 degrees phase difference. The measured maxima are 0% and 0 degrees.

Baseline equivalence

Evidence	Acceptance	Measured	Result
Relative RMS error, 9 signals	$\leq 1e-8$	0	Pass
Normalized maximum error	$\leq 1e-8$	0	Pass
Model-file byte identity	Required	Identical	Pass

The verified channels are Vd, Vq, Id, Iq, active power, reactive power, PLL frequency, SoC and fault state.

4. Grid interaction metric

The grid is represented by a synchronous-frame Thevenin impedance for SCR values 10, 5, 3 and 2. Converter rating and grid X/R ratio remain fixed. The interaction matrix and screening score are:

$$L(j\omega) = Z_{grid}(j\omega) Y_{bess}(j\omega)$$
$$\gamma(w) = \text{maximum singular value of } L(j\omega)$$

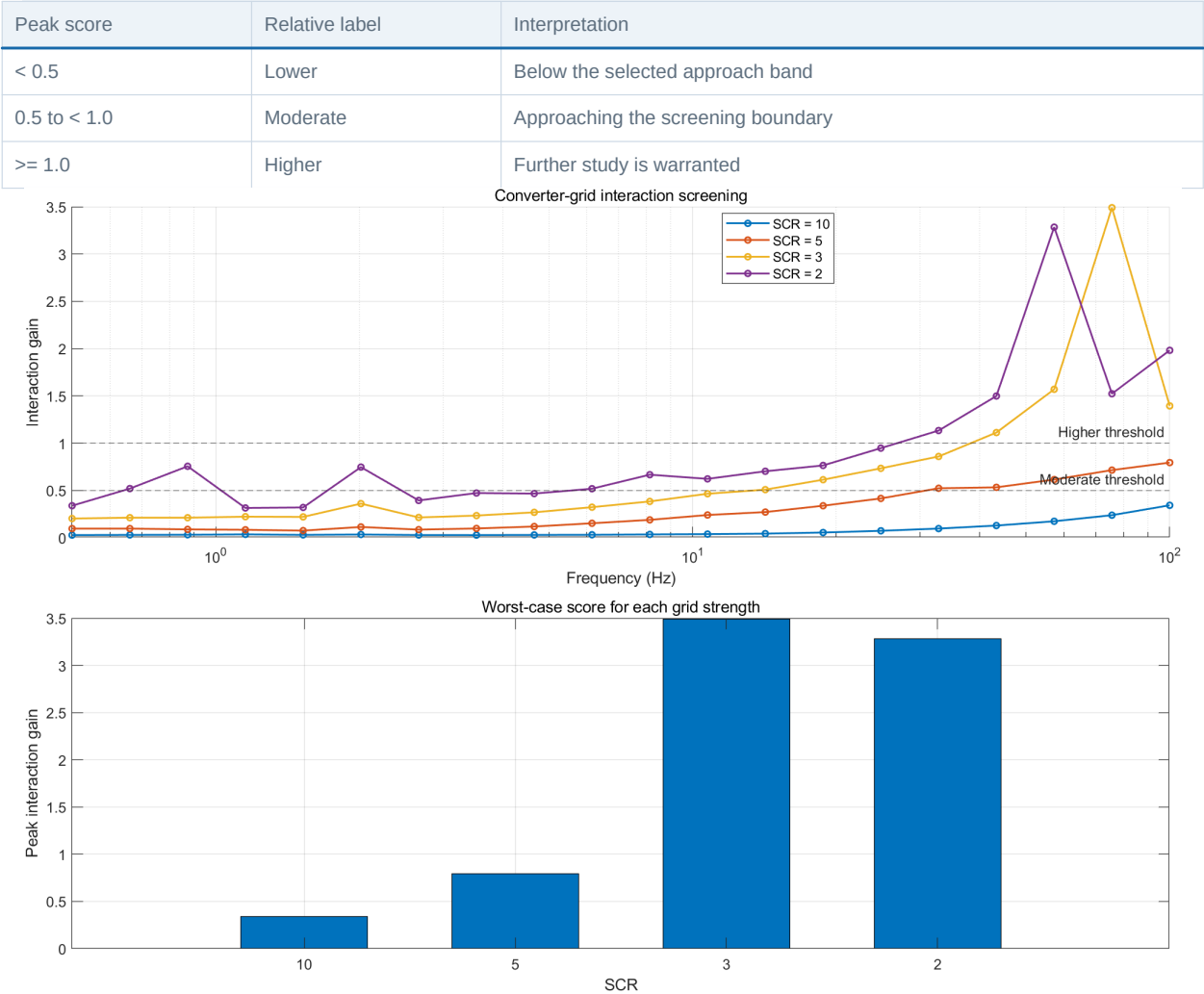


Figure 4. Interaction score for four grid strengths at nominal PLL scale.

Nominal-PLL peaks are 0.343 at SCR 10, 0.795 at SCR 5, 3.490 at SCR 3 and 3.284 at SCR 2. The critical frequency shifts from 100 Hz to 75.66 Hz and then 57.25 Hz as the grid weakens.

The labels are fixed relative screening categories. They do not declare an operating point formally stable or unstable.

5. SCR-PLL risk map

PLL proportional and integral gains are scaled together by 0.5, 1.0, 1.5 and 2.0. All sixteen cases use the same frequency vector, identification method, quality limits and risk thresholds.

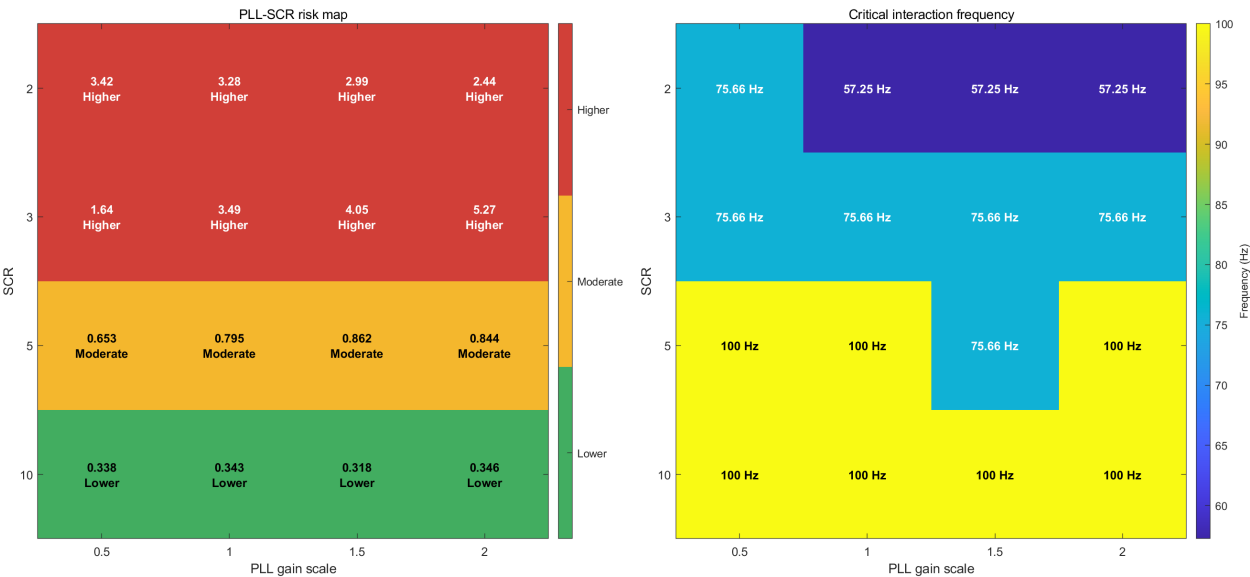


Figure 5. Peak interaction score, relative label and critical frequency for each SCR-PLL pair.

SCR / PLL	0.5	1.0	1.5	2.0
10	0.338 L	0.343 L	0.318 L	0.346 L
5	0.653 M	0.795 M	0.862 M	0.844 M
3	1.643 H	3.490 H	4.055 H	5.270 H
2	3.418 H	3.284 H	2.991 H	2.442 H

The strongest trend is the score increase as the grid weakens from SCR 10 to SCR 3. PLL sensitivity also increases, but its direction is not universal: score rises with PLL scale at SCR 3 and falls at SCR 2. The full matrix is therefore more informative than a single rule about PLL bandwidth.

6. Time-domain cross-check

Three cases receive the same active-power reference step. The response is normalized and evaluated through overshoot, 5% settling, dominant oscillation frequency and the late-to-early oscillation RMS ratio.

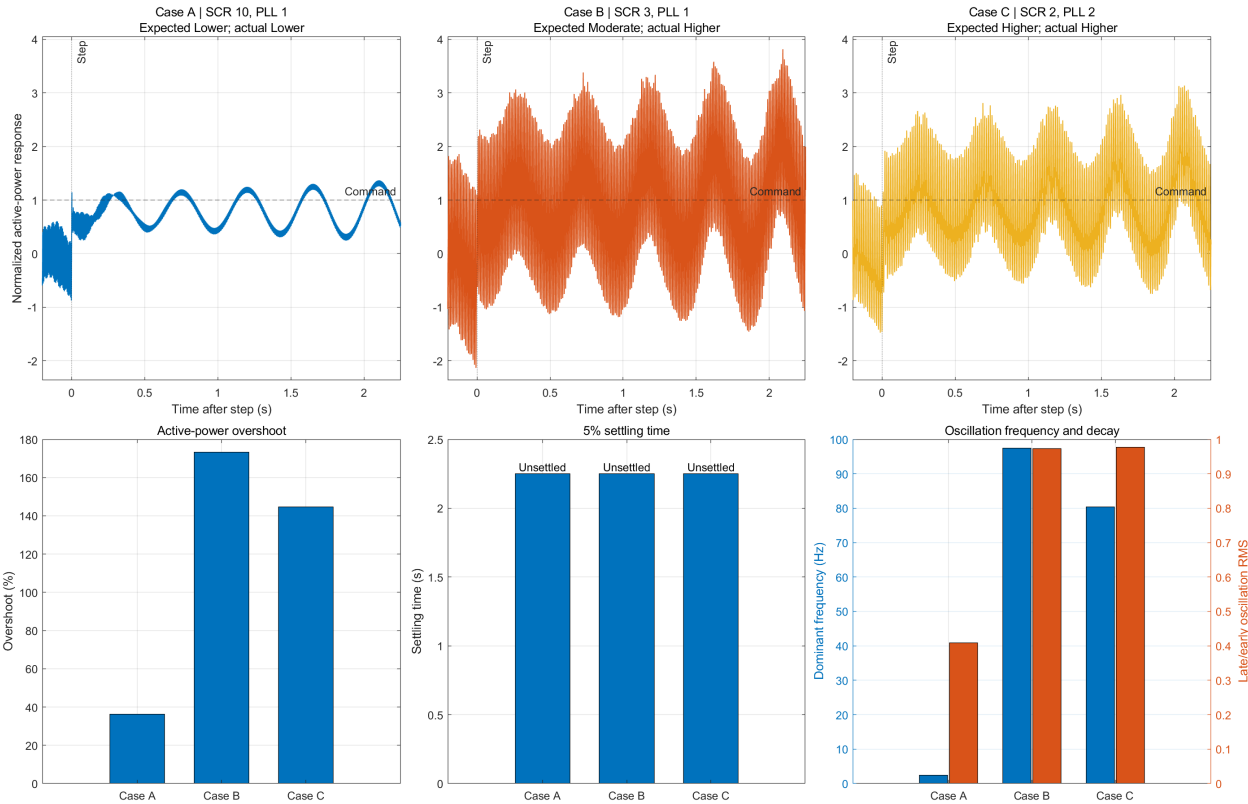


Figure 6. Representative time-domain responses and comparison metrics.

Case	SCR	PLL	Planned / actual	Overshoot	Frequency	Decay
A	10	1.0	Lower / Lower	36.15%	2.437 Hz	0.4090
B	3	1.0	Moderate / Higher	173.3%	97.48 Hz	0.9733
C	2	2.0	Higher / Higher	144.7%	80.42 Hz	0.9778

All three cases remain outside the 5% band after 2.25 s, so settling-time ordering is not claimed. Overshoot ordering agrees with frequency-domain ordering for 3/3 comparable pairs; decay-ratio ordering agrees for 2/3.

Case B is retained as calculated. Its Higher label differs from the planned Moderate role and is treated as an engineering finding rather than a software failure.

7. Reproduction, evidence and limitations

Run the complete local validation from the project folder:

```
cd('path/to/bess-frequency-scan-stability-screening')
run('tests/run_all_checks.m')
```

Verification evidence

Check group	Evidence
Single-frequency interface	Independent d/q voltage and current response
Repeatability	Two separate 10 Hz runs; magnitude and phase comparison
Baseline equivalence	Nine signals plus model-file identity
Nominal sweep	20 frequencies; 40 injection runs
SCR screening	Four grid strengths; critical frequency
PLL risk map	Sixteen operating cases; 640 represented runs
Time-domain check	Three cases; overshoot, settling, frequency and decay

Limitations

- Average-converter simulation is used instead of switching EMT or hardware measurement.
- Admittance is local to the selected operating point and perturbation amplitude.
- The singular-value score is comparative screening, not a formal generalized Nyquist proof.
- Only SCR and PLL scale are swept; X/R, P/Q point, delay and uncertainty are not full dimensions.
- Twenty frequency points locate broad regions but cannot guarantee the continuous-frequency maximum.
- The time-domain window is finite and all representative cases are unsettled at its end.

Conclusion

Within these boundaries, the workflow meets its purpose: repeatable terminal perturbations produce a complete dq response, one fixed metric compares weak-grid and PLL sensitivity, critical frequency regions are retained, and mismatches remain visible for engineering interpretation.